MD3135 DATA SCIENCE COURSE PROJECT

Project Proposal & Expected Outcome Format

Third Year – Data Science Course

Department	Computer Engineering
Academic Year	2026-27
Course/Subject	MD3135: Data Science
Project / Team ID	MDM-DS-
Project Guide	Dr. Kapil Vhatkar

1. STUDENT / TEAM DETAILS Each group must consist of a minimum of 1 student and a maximum of 4 students.

Sr. No.	Student Name	Div	Tutorial Batch	Roll No.	Email ID
1	Karkele Om Vivekanand	F	B13	31	om.karkele24@vit.edu
2					
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2. PROJECT IDENTIFICATION

1. Project Title *Write a clear and meaningful title.*

MLOps-Integrated Fraud Detection with Drift-Aware Retraining: A Comparative Study of Retraining Policies under Concept Drift

2. Project Area / Domain *Examples: Healthcare, Education, Finance, Agriculture, E-Commerce, Sports, Environment, etc.*

FinTech, Fraud Detection, MLOps

3. Problem Statement *What problem are you trying to solve using Data Science?*

Credit card fraud detection models can experience performance degradation over time because fraud patterns and cardholder behavior continuously evolve, resulting in concept drift. Most fraud detection studies evaluate models using static train-test splits and do not adequately examine how model performance changes after deployment or when the model should be retrained. This project aims to simulate a chronological deployment environment, detect concept drift in fraud patterns, and compare different retraining strategies - no retraining, fixed-interval retraining, and drift-triggered retraining based on predictive performance and operational cost.

4. Aim of the Project *State the overall purpose in 1–2 sentences.*

To develop a fraud detection system that can identify and adapt to changing fraud patterns using concept drift detection and automated model retraining. The project aims to compare different retraining strategies based on predictive performance and operational cost.

5. Objectives *List 3–5 specific objectives.*

1. To build a fraud detection model using transaction data and evaluate its performance on data arranged in chronological order.
2. To identify changes in fraud patterns over time using suitable concept drift detection techniques.
3. To compare different model retraining strategies, including no retraining, fixed-interval retraining, and drift-triggered retraining.
4. To evaluate these strategies based on fraud detection performance and the operational cost of retraining.
5. To use model explainability techniques to understand the factors influencing fraud predictions.

3. DATASET AND DATA UNDERSTANDING

6. Dataset / Data Source *Mention dataset name, source/link and how the data will be obtained.*

IEEE-CIS Fraud Detection Dataset, obtained from Kaggle. The dataset contains transaction and identity-related information for detecting fraudulent transactions. The data will be downloaded and processed locally using Python. The transactions will be arranged in chronological order and divided into sequential time windows to simulate a real-world fraud detection environment.
Link: https://www.kaggle.com/competitions/ieee-fraud-detection/data

7. Important Features / Variables *Mention important attributes/features and the target variable, if applicable.*

Important transaction features include TransactionID, TransactionDT, ProductCD, card1-card6, addr1, addr2, email-domain features, and M1-M9. Identity-related features include DeviceType, DeviceInfo, and id_12-id_38. The transaction and identity data will be joined using TransactionID. The target variable is **isFraud**, which indicates whether a transaction is fraudulent (1) or non-fraudulent (0).

4. DATA SCIENCE METHODOLOGY

8. Proposed Methodology *Explain the planned workflow: Data Collection → Cleaning → EDA → Visualization → Analysis/Model → Evaluation → Insights.*

The project will begin by collecting and understanding the IEEE-CIS Fraud Detection dataset, followed by cleaning the transaction and identity data and handling missing and inconsistent values. Exploratory Data Analysis (EDA) and visualizations will be used to understand transaction patterns, class imbalance, and changes in fraud behavior over time. The cleaned data will then be arranged chronologically and used to build a baseline fraud detection model. Concept drift will be monitored using suitable drift detection techniques, and three retraining strategies no retraining, fixed-interval retraining, and drift-triggered retraining will be compared. The models will be evaluated using suitable performance and cost-based measures, and the results will be analyzed to identify the most effective retraining approach and the factors influencing fraud predictions.

9. Data Preprocessing Techniques *Mention techniques such as missing-value handling, outlier treatment, encoding, scaling, feature selection, etc.*

Missing values will be identified and handled appropriately based on the nature of each feature. Categorical variables will be encoded into a suitable numerical format, while numerical features will be checked for inconsistent values and extreme observations. Relevant features will be selected based on their usefulness for fraud detection, and unnecessary or highly redundant features will be removed. Class imbalance between fraudulent and non-fraudulent transactions will be addressed using techniques such as class weighting or SMOTE. The data will then be arranged chronologically to preserve the time-based nature of the problem and avoid data leakage.

10. Data Analysis / Machine Learning Techniques *Examples: descriptive statistics, correlation, regression, classification, clustering, time-series, etc.*

Descriptive statistics and exploratory analysis will be used to understand transaction patterns, fraud distribution, and relationships between important features. Binary classification models such as XGBoost, LightGBM, or CatBoost will be used for fraud detection. Concept drift detection techniques such as ADWIN and DDM will be applied to identify changes in model performance over time. SHAP will be used to understand the features contributing to fraud predictions.

11. Tools / Technologies *Examples: Python, Pandas, NumPy, Matplotlib, Seaborn, Scikit-learn, Jupyter/Colab, Streamlit.*

Python, Pandas, NumPy, Scikit-learn, XGBoost/LightGBM/CatBoost, River, Imbalanced-learn, Matplotlib/Plotly, SHAP, Jupyter Notebook, MLflow, Docker, and GitHub Actions.

5. EXPECTED RESULTS

12. Expected Outcome *What do you expect to discover, predict, classify, analyze or visualize?*

The project is expected to develop a fraud detection model and analyze how its performance changes as fraud patterns evolve over time. It will compare no-retraining, fixed-interval retraining, and drift-triggered retraining to identify which approach provides better and more consistent fraud detection performance. The project will also provide visualizations of model performance and drift over time and use explainability techniques to understand the factors influencing fraud predictions.

13. Key Insights Expected *Mention the useful insights or decisions expected from the analysis.*

The analysis is expected to provide insights into how fraud patterns and model performance change over time. It will help identify whether concept drift significantly affects fraud detection performance and determine which retraining strategy provides a better balance between detection performance and retraining cost. The analysis will also help identify the important features that contribute to fraud predictions and provide a basis for selecting an appropriate model maintenance strategy.

6. EXPECTED PROJECT DELIVERABLES

Select the deliverables that are applicable:

- ☑ Dataset / Data Collection
- ☑ Cleaned Dataset
- ☑ Python / Jupyter Notebook
- ☑ Exploratory Data Analysis (EDA)
- ☑ Data Visualizations / Dashboard
- ☑ Statistical Analysis
- ☑ Machine Learning Model
- ☑ Prediction / Classification / Clustering Results
- ☐ Working Software / Application
- ☑ Project Report
- ☑ Presentation / Demonstration
- ☐ Other: _____

7. EXPECTED RESEARCH / PROFESSIONAL OUTCOME

Select the expected outcome after successful completion of the project:

- ☑ Course Project Report
- ☐ Book Chapter
- ☐ Scopus-indexed Conference Publication
- ☐ Scopus/SCI / SCIE / ESCI Journal Publication
- ☐ Patent Application

☐ Review / Survey Paper

☑ Poster / Technical Demonstration

☑ Research Dataset / Benchmark Dataset

☑ Dashboard / Application

☑ Open-Source GitHub Contribution

☐ Product / Commercialization Opportunity

☐ Startup / Entrepreneurship Opportunity

☐ Startup / Entrepreneurship Opportunity

☐ Other: _____

8. REFERENCES

14. References / URLs *Mention at least 3 important references, research papers, datasets or existing systems.*

1. Adebayo, O.S. et al. (2023). *Comparative Review of Credit Card Fraud Detection using Machine Learning and Concept Drift Techniques.* IJCSMC, Vol. 12, Issue 7.
2. Dal Pozzolo, A., Boracchi, G., Caelen, O., Alippi, C., & Bontempi, G. (2018). *Credit Card Fraud Detection and Concept-Drift Adaptation with Delayed Supervised Information.* IEEE Transactions on Neural Networks and Learning Systems.
3. Bifet, A. & Gavaldà, R. (2007). *Learning from Time-Changing Data with Adaptive Windowing.* Proceedings of the SIAM International Conference on Data Mining.
4. Gama, J., Medas, P., Castillo, G., & Rodrigues, P. (2004). *Learning with Drift Detection.* Brazilian Symposium on Artificial Intelligence.
5. IEEE Computational Intelligence Society. *IEEE-CIS Fraud Detection Dataset.* Kaggle.

9. STUDENT DECLARATION

We confirm that the information provided in this project proposal is correct to the best of our knowledge. We understand that the project should demonstrate the application of Data Science concepts, including data preparation, exploratory analysis, visualization and appropriate analytical or machine-learning techniques.

Team Leader	Team Member 2	Team Member 3	Team Member 4
Signature: Omkarkele Date: August 25, 2026	Signature: _____ Date: _____	Signature: _____ Date: _____	Signature: _____ Date: _____

10. FOR FACULTY USE

Evaluation Parameter	Remarks	Status
Problem Relevance		☐ Approved ☐ Revise
Dataset Suitability		☐ Approved ☐ Revise
Data Science Methodology		☐ Approved ☐ Revise
Technical Feasibility		☐ Approved ☐ Revise
Expected Outcome		☐ Approved ☐ Revise
Overall Scope		☐ Approved ☐ Revise